

Covid 19 and its impact on students

Mohamed Salah Salem
Dept. of Information Systems
College of IT, University of Bahrain
Sakhir, Kingdom of Bahrain
202206016@stu.uob.edu.bh

Ammar Yasser
Dept. of Information Systems
College of IT, University of Bahrain
Sakhir, Kingdom of Bahrain
202202885@stu.uob.edu.bh

Mohamed Saleh salem
Dept. of Information Systems
College of IT, University of Bahrain
Sakhir, Kingdom of Bahrain
202203322@stu.uob.edu.bh

Salman Sabah
Dept. of Information Systems
College of IT, University of Bahrain
Sakhir, Kingdom of Bahrain
202004607@stu.uob.edu.bh

Supervised by: Dr. Ahmed Zeki

Abstract

The COVID-19 crisis shattered students' routines and forced a switch to online courses, which created time management problems. This paper examines 1,182 Delhi NCR student survey responses, investigating the effect of study time, fitness time, sleep time, and media usage time on perceived time use. From J48 and Naïve Bayes models, we predicted whether the time used was optimally. Naïve Bayes worked marginally better than J48 (61.6% compared to 61.0%). Key predictors included fitness, self-study, and social media control, highlighting good habits in online learning.

I. INTRODUCTION

A. Background:

COVID-19 lockdowns steered education in the home environment, wherein the children had to self-learn time management. Screen usage skyrocketed through 11 hours of their weekly usage, which disrupted sleep, reduced concentration, and resulted in poorer grades. Most of them were unable to cope with timetabling as much as classes, homework, and sleeping without a human body.

B. Problem Description:

Students reported difficulty managing time, missing deadlines, and spending too much on non-academic activities. Others coped by using schedules or family support. This study analyzes student activity patterns class, study, fitness, sleep, media—and predicts if their time was “utilized” (Yes/No). Our goal is to identify habits linked to effective time use during online learning.

II. DATA COLLECTION

For this study, we utilized a publicly available dataset sourced from Kaggle, titled COVID-19 Survey Student Responses, which was originally compiled by researchers from Delhi, India, at the beginning of the pandemic[1]. The dataset contains 1,182 instances from students spanning schools in the Delhi National Capital Region (NCR). It records daily time spent on Online

perception whether they had used their time effectively. The dataset was downloaded from Kaggle in ARFF format, preprocessed and ready to be used in data mining software.

A. Similar Studies

Other global research recorded similar student time management patterns during the COVID-19 pandemic. A study conducted in Malaysia, for example, observed that the majority of undergraduates were not effective time managers, although female students performed better than male students[2]. Additionally, students with consistent study habits in an online programming course were found to be more engaged and performed better. These findings validate a common pattern: good time management was a significant predictor of academic success within virtual learning environments. Our study builds on this foundation by applying machine learning techniques to identify which specific daily activities have the greatest impact on students' self-reported use of time.

B. Objectives

In this article, an effort has been made to explore how students managed time during lockdown in the COVID-19 pandemic, predict the efficient use of time using Decision Tree and Naïve Bayes classification algorithms, and identify influential behavior factors namely, study time and use of social media which affect time management.

III. DEVELOPMENT STAGES

A. Data Preparation

Data required minimal cleaning because it was already preformatted with six numeric activity features and one nominal class feature ("Time utilized": YES/NO). We searched for missing values, maintained uniform numeric formats, and normalized outliers. Some demographic columns were dropped while preprocessing, and only relevant time-based features survived. Class distribution was balanced (51.3% NO, 48.7% YES), and accuracy was hence an appropriate metric. The data set was now ready to be trained on by models with cross-validation.

Class, Self Study, Fitness, Sleep, Social Media, and TV, with the target attribute "Time utilized" (Yes/No), i.e., students'

TABLE I. ATTRIBUTES

Attribute	Description	Value
Time spent on Online Class	Time (in hours) spent attending online classes	Numeric (e.g., 0.0, 2.0, 5.0, ...)
Time spent on self study	Time (in hours) spent on self-study or academic preparation	Numeric (e.g., 0.0, 3.0, 4.0, ...)
Time spent on fitness	Time (in hours) spent on physical activity or fitness	Numeric (e.g., 0.0, 1.0, 2.0, ...)
Time spent on sleep	Time (in hours) spent sleeping	Numeric (e.g., 5.0, 8.0, 10.0, ...)
Time spent on social media	Time (in hours) spent on social media platforms	Numeric (e.g., 1.0, 3.0, 6.0, ...)
Time spent on TV	Time (in hours) spent watching television or streaming	Numeric (e.g., 0.0, 2.0, 5.0, ...)
Time utilized	Whether the person's time is considered utilized productively	Nominal: YES / NO

B. J48 Algorithm

We chose the J48 decision tree for its clear, interpretable structure and effectiveness in classification. To ensure reliable results, we evaluated the model using 10-fold cross-validation.

Correctly Classified Instances	721	60.9983 %
Incorrectly Classified Instances	461	39.0017 %
Kappa statistic	0.2225	
Mean absolute error	0.4579	
Root mean squared error	0.4986	
Relative absolute error	91.6604 %	
Root relative squared error	99.7549 %	
Total Number of Instances	1182	

Fig. 1. (J48) Summary of J48 stratified cross-validation results.

As Figure 1 illustrates, the model correctly classified 721 instances out of 1182, or 60.99% accuracy. Meanwhile, 461 instances (or 39.00%) were incorrectly classified. This indicates that the J48 algorithm was able to identify important patterns in the data for student use of time.

=== Detailed Accuracy By Class ===									
	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.671	0.447	0.586	0.671	0.626	0.225	0.610	0.574	YES
	0.553	0.329	0.640	0.553	0.593	0.225	0.610	0.589	NO
Weighted Avg.	0.610	0.387	0.614	0.610	0.609	0.225	0.610	0.582	

Fig. 2. (J48) detailed accuracy by class.

This is the classification performance of the J48 model for both "YES" and "NO" classes. Some key evaluation measures include True Positive Rate (TP Rate), False Positive Rate (FP Rate), Precision, Recall, and F-Measure. The model provides a TP Rate of 0.671 and Recall of 0.671 for the "YES" class, and for the "NO" class, the TP Rate and Recall are 0.553. Overall Precision, Recall, and F-Measure weighted averages are 0.614, 0.610, and 0.609 respectively. Class-wise ROC Area is 0.610, indicating moderate discriminatory power. All these measures indicate relatively well-balanced performance for both outcome classes.

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=== Confusion Matrix ===

      a    b    <-- classified as
385 189 |    a = YES
272 336 |    b = NO

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Fig. 3. (J48) confusion matrix.

This confusion matrix embodies both class "YES" (positive instances) and class "NO" (negative instances) classification results. The model assigned 385 to "YES" correctly and 336 to "NO" correctly. However, 189 "YES" were mistakenly counted as "NO," while 272 instances of "NO" were miscounted as "YES." That is, the model was performing better to identify "YES" cases than "NO" cases but might have to cut down misclassification.

C. Naïve Bayes Algorithm

We used the Naïve Bayes classifier in our study as an additional method to support and validate our findings. Its probabilistic approach is well-suited for predicting diabetes. We used 10-fold cross-validation again for its consistency and fair comparison.

Correctly Classified Instances	728	61.5905 %
Incorrectly Classified Instances	454	38.4095 %
Kappa statistic	0.2328	
Mean absolute error	0.4526	
Root mean squared error	0.4916	
Relative absolute error	90.5902 %	
Root relative squared error	98.3542 %	
Total Number of Instances	1182	

Fig. 4. (Naïve Bayes) summary of stratified cross-validation results.

As shown in Figure 4, the Naïve Bayes classifier correctly predicted 728 out of 1182 instances and attained an accuracy level of 61.59%. Furthermore, 454 instances were wrongly predicted and account for 38.41% of the instances. These values are a moderate performance and signify that the model obtained suitable patterns in student behavior up to the utilization of time.

=== Detailed Accuracy By Class ===									
	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.643	0.410	0.597	0.643	0.619	0.233	0.647	0.600	YES
	0.590	0.357	0.637	0.590	0.613	0.233	0.647	0.639	NO
Weighted Avg.	0.616	0.383	0.617	0.616	0.616	0.233	0.647	0.620	

Fig. 5. (Naïve Bayes) detailed accuracy by class.

This graph is a comparison of how well the Naïve Bayes classifier performed on the "YES" and "NO" classes. The model achieved a TP Rate of 0.643 and an F-Measure of 0.619 for the "YES" class, and a TP Rate of 0.590 with an F-Measure of 0.613 for the "NO" class. The weighted averages were Precision: 0.617, Recall: 0.616, and F-Measure: 0.616. Both classes scored a ROC Area of 0.647, indicating a moderate ability to distinguish between productive and unproductive time use. Overall, the model performed slightly better at predicting the "YES" class in terms of recall and F-Measure, while showing higher precision for the "NO" class.

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=== Confusion Matrix ===
      a      b  <-- classified as
369 205 |      a = YES
249 359 |      b = NO
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Fig. 6. (Naïve Bayes) confusion matrix

Confusion matrix shows performance of the classification made by Naïve Bayes model. It correctly classified 369/369 instances belonging to the class "YES" and 359/359 instances belonging to the class "NO". However, 205 samples of "YES" were wrongly labeled as "NO", and 249 samples of "NO" were wrongfully labeled as "YES". The result points toward a decently balanced classification of both the classes with minimal dominance of class "YES" being recognized over the other one.

IV. RESULTS

A. J48 Decision Tree

The J48 decision tree model classified the students on the basis of whether they made efficient use of their time amid the COVID-19 pandemic. With 10-fold cross-validation, the model accuracy was found to be 61.6%, indicating moderate predictive power. The algorithm found primary behavioral features—time spent on fitness, self-study, sleep, and online lessons—as key predictors of academic time management. These characteristics were consolidated into a structured decision tree, whereby the model is made understandable and conducive for deriving useful practical rules. The entire decision tree structure formed utilizing the J48 algorithm is presented below in Figure 7.

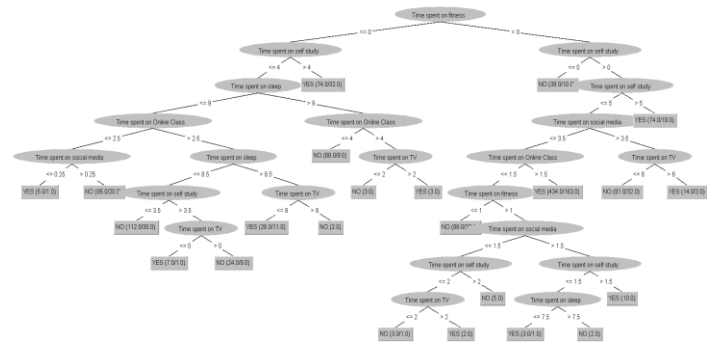


Fig. 7. Decision tree representation.

The decision tree in Figure 7 illustrates the J48 algorithm classifying students based on their usage of time during the COVID-19 lockdown. The tree begins with "Time spent on fitness," with huge splits such as "Time spent on self study," "Time spent on Online Class," and "Time spent on sleep" and "TV." Each of these splits reveals behavioral thresholds leading to a classification as "YES" (good usage of time) or "NO".

For instance, those who had zero hours of fitness time and fewer hours of self-study were largely classified as "NO". Conversely, those who worked for more hours on online classes, self-study, and limited their time on TV and social media were leaning towards the "YES" category. The tree puts forward the investment of equal hours in study and non-study activities, most importantly self-study and structured entertainment, as the essence of effective time management. The visualization provides self-evident and practical information on student behavioral tendencies.

B. Naïve Bayes Classifier

Apart from the J48 algorithm, we applied the Naïve Bayes classifier to validate our observation of the time management of students during the COVID-19 lockdown. This model was 61.6% accurate, marginally more than J48 in this study. Naïve Bayes relies on probabilistic assumptions and is particularly well known for performance with datasets consisting of independent features. As shown by the stratified cross-validation report (Figure 4), the model made 728 out of 1182 instances correct. The detailed accuracy measures (Figure 5) identify the model to be very better in predicting the "NO" (ineffective time management) class, even though it also maintains a good balance in predicting "YES" outcomes. Naïve Bayes is a simple but competitive classifier for behavior pattern analysis in this case as a whole.

C. Insights and Actionable Outcomes

- If Time Spent on Fitness = 0 and Time Spent on Self-Study > 4, the model predicts "YES" (effective time utilization).
This indicates that even without physical activity, students who invest more time in self-study are more likely to manage their time effectively.
- If Time Spent on Fitness = 0, Time Spent on Self-Study ≤ 4, and Rating of Online Class Experience = Good, the outcome depends on sleep and social media time:
- If Time Spent on Sleep ≤ 9 and Self-Study > 1.5 and Social Media ≤ 4 → "YES"
- If Time Spent on Sleep > 9 or Social Media > 4 → "NO"
This suggests that a positive class experience and moderate social media use contribute to better time management.
- If Time Spent on Fitness > 0 and Online Class Time > 1.5, the model predicts "YES", regardless of other factors.
This emphasizes the value of physical activity and online class engagement in promoting effective time use.
- If Fitness > 0, Online Class ≤ 1.5, and Self-Study ≤ 2, the outcome depends on TV and Sleep Time:
- If Time on TV ≤ 2 → "NO"
- If Time on TV > 2 and Sleep > 7.5 → "NO"
- Otherwise → "YES"
These cases suggest that poor time use can stem from lack of study time and high screen time, even if physical activity is present.
- If Self-Study ≤ 0, the model predicts "NO", regardless of other activities.
- → This reinforces that self-study is a crucial determinant of time management success.

These decision rules highlight how combinations of academic effort, screen time, and lifestyle habits like fitness and sleep interact to influence students' ability to manage their time effectively during remote learning. Let me know if you'd like this included in the full report layout.

V. CONCLUSION

The combination of J48 and Naïve Bayes enabled cross-validation of our findings and combining the findings of the study. Although the J48 decision tree produced a comprehensible linear model for the identification of primary behavioral patterns, the Naïve Bayes classifier offered solid probabilistic affirmation. The two models agreed in highlighting the role of online course participation, use of social media, and self-study hours as predictors of productive time usage by students. These results can be employed to inform interventions to facilitate future academic productivity, especially in online or blended courses.

CONTRIBUTION SHEET

Meeting	Date	Discussion	Key Contributions
Meeting 1: Initial Planning	25-Feb	Established the project objectives, delegated team responsibilities, and chose appropriate datasets for analysis	- Mohammed Saleh : Proposed the project topic. - Mohammed Salah : Suggested dataset options & discussed feature selection. - Salman Subah : Outlined the report structure. - Ammar Yasser : Took meeting notes and organized tasks.
Meeting 2: Preprocessing the Dataset	10-Mar	Reviewed the dataset and applied preprocessing techniques (e.g., handling missing values, transforming data).	- Mohammed Saleh : Documented the preprocessing steps. - Mohammed Salah : clearing the data - Salman Subah : Applied transformation methods. - Ammar Yasser : Documented preprocessing steps.
Meeting 3: Algorithm Implementation	16-17 Apr	Implemented J48 and Naive Bayes classifiers and evaluated model performance.	- Mohammed Saleh : Ran the J48 algorithm and interpreted the decision tree. - Salman Subah : Implemented and tested Naive Bayes. - Mohammed Saleh : Summarized Naive Bayes results. - Ammar Yasser : Summarized J48 results.
Meeting 4: Writing the Report	27-Apr	Worked together to draft the report, covering the problem statement, methodology, results, and final conclusions	- Mohammed Saleh : Reviewed the conclusion and contribution sheet. - Mohammed Salah : Drafted the results section and discussed J48 findings. - Salman Subah : Designed layout and worked on formatting. - Ammar Yasser : Reviewed and finalized the report.
Meeting 5: Final Review and Submission	3-May	Performed a comprehensive final review to ensure all deliverables were complete and ready for submission	All members verified.

REFERENCES

- [1] Kunal Chaturvedi, "COVID-19 and Its Impact on Students," Kaggle. [Online]. Available: <https://www.kaggle.com/datasets/kunal28chaturvedi/covid19-and-its-impact-on-students>. [Accessed: May 2025].
- [2] Nesamalar, J., Tan, P. L., & Singaram, N. (2022). *Time Management Behaviour During the COVID-19 Pandemic: A Focus on Higher Education Students*. Asia-Pacific Journal of Futures in Education and Society, 1(1), 17–38. Available at: <https://www.researchgate.net/publication/360081404>